

# Origami Circuits: How Folding Paper Can Power Lights and Motors

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Imagine folding an origami crane and, with a few extra creases, making its wings light up. That's the promise of a new material developed by Qi Zhang and Kening Zhu at the City University of Hong Kong. Their team has created a paper-like nonwoven textile impregnated with liquid metal ink that transforms into a working circuit when you fold or cut it. The magic lies in the physics of the ink. Initially, the material is **nonconductive** because an insulating **oxide layer** forms around microscopic droplets of a **gallium-indium** liquid metal. Applying pressures as little as **2.5 megapascals** from a simple fold cracks that layer, allowing the droplets to merge and create conductive traces. "We confirmed that normal handling of the paper didn't accidentally create conductive paths," says Zhu, ensuring that only intentional folds activate the circuit. To make the material accessible, the team optimized a process using an airbrush to spray the ink mixture, allowing **stencils** to mask areas that should remain **nonconductive**. The result: folded edges have a **resistance** of just 4.4 ohms per centimeter, and cut edges 23.18 ohms per centimeter. Electronic components like LEDs and motors are attached with conductive thread and masking tape. The researchers created a toolkit called LiqMetCraft, which includes a browser-based design app. Users can choose from three papercraft styles: Chinese radial papercutting, origami with 12 templates, or 3D model making. The software provides step-by-step guidance and even interfaces with a cutting machine. At about US \$1.80 per 10-by-10-centimeter piece, this innovation could revolutionize STEM education, letting students literally fold their way into electronics.

## Word Treasure

**nonconductive** -- Not allowing electricity to flow through; an insulator before pressure is applied.

**oxide layer** -- A very thin coating that forms on some metals when exposed to air, acting as an electrical insulator.

**megapascals** -- A unit of pressure; 2.5 MPa is about the force you'd apply with a firm finger press.

**gallium-indium** -- A metal alloy that stays liquid at room temperature and is safe to use in soft electronics.

**stencils** -- Cut-out patterns that let ink pass through only in certain shapes, used to control where the circuit forms.

**resistance** -- A measure of how much a material fights the flow of electricity, measured in ohms.

## Background you need

Origami, the Japanese art of paper folding, has inspired engineers for decades. In space exploration, folding solar panels and sunshields use origami principles to launch compactly and expand in orbit. Soft robotics and wearable electronics have also explored conductive fabrics, but often require complex wiring. The field of liquid metal electronics is not new: gallium-based alloys, which are liquid at room temperature and can form a protective oxide skin, have been used in stretchable circuits. Previous researchers at Carnegie Mellon University and others have developed liquid metal-injected fibers. The team at the City University of Hong Kong (a public university with a focus on technology) built a simple airbrush-based process to make this material affordable. Their LiqMetCraft toolkit could bring foldable electronics to schools, combining art and engineering. STEM education (science, technology, engineering, and math) aims to equip students with problem-solving skills for future careers.

## Break it down

**LEAD:** Origami crane lighting up with few extra creases

**MATERIAL:** Paper-like nonwoven textile with liquid metal ink

+ INITIAL STATE: Nonconductive due to oxide layer

**ACTIVATION PHYSICS:** Pressure (2.5 MPa from folding) cracks oxide, merges droplets

+ RESISTANCE: Folded: 4.4  $\Omega$ /cm; cut: 23.18  $\Omega$ /cm

**FABRICATION:** Airbrush stencil-masked ink, attach components with conductive thread

**TOOLKIT:** LiqMetCraft (browser-based design app + papercraft styles)

+ STYLES: Chinese radial, origami (12 templates), 3D model

+ AUTOMATION: Interfaces with cutting machine

**COST & POTENTIAL:** ~\$1.80 per piece, aim to revolutionize STEM education

## Why it matters

This material could democratize electronics, letting anyone turn a simple fold into a working light or motor, which might inspire more students to pursue careers in engineering and design, blending art and technology to solve future challenges.

## Test what you remember

1. What material is used to create the paper-like circuit?
  - (A) Copper wire.
  - (B) Nonwoven textile with liquid metal ink.
  - (C) Graphene sheets.
  - (D) Plastic film with silver paint.
2. What causes the material to become conductive after folding?
  - (A) Heat from fingers melts the metal.
  - (B) Pressure cracks the insulating oxide layer, merging droplets.
  - (C) Light activates a chemical reaction.
  - (D) Static electricity aligns particles.
3. What is the resistance of a folded edge in the material?
  - (A) 2.5 megapascals.
  - (B) 4.4 ohms per centimeter.
  - (C) 23.18 ohms per centimeter.
  - (D) 1.80 ohms per meter.
4. How is the liquid metal ink applied to the material?
  - (A) By dipping in a bath.
  - (B) With a roller.
  - (C) Using an airbrush with stencils.
  - (D) Printed by an inkjet printer.
5. What does the LiqMetCraft toolkit include?
  - (A) A 3D printer and soldering iron.
  - (B) A browser-based design app and papercraft templates.
  - (C) A set of pre-cut circuit boards.
  - (D) A chemistry lab kit.
6. What is the approximate cost of a 10-by-10-centimeter piece of the material?
  - (A) US \$0.50.
  - (B) US \$1.80.
  - (C) US \$10.00.
  - (D) US \$4.40.

Answer key (try the quiz first!): 1: B 2: B 3: B 4: C 5: B 6: B

